

Mine in a box™

Managing Operations

Part A · Supervisor Apply Tools

Systems that make good decisions inevitable.

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How to use these tools

You get these tools with the Supervisor Handbook at the Validation Lab. Use them on every shift for the 6 to 8 weeks of Apply. Your Superintendent coaches you from them. The same tools will be in Sam.

Tool	What it is	When
1	Work out your number	Before the pre-start
2	The operator's count card	Every block, by each operator
3	The operator scorecard	Each roster, on the wall
4	The handover sheet and cards (and the pre-start card)	Both handovers, and the pre-start
5	The four block cards	Every shift, end of each block
6	The Shift Report	Through the blocks; done by about 16:00
7	The assumptions I inherited	Redo each quarter
8	The six behaviours self-evaluation	Each roster
9	The fault-tree tool	When a loss keeps coming back

PAVA means Plan, Actual, Variance/Impact, Actions taken. It is how every number is reported.

Card lines. Each card line has the same number as in the handbook (line 1 to line 44, and "+" on every card). The number is the line's ID in Sam. Mark each line Yes or No. Every No gets an action, an owner and a date.

Not available yet at my site. If a figure, a system or a person is not available, write "not available yet" and tell your Superintendent. If a step needs your Superintendent and they are not there, do the step, write it down, and take it to them at the next meeting.

Times are indicative. Your site can change them. The order does not change.

Tool 1. Work out your number

Before the pre-start

Fill it in from your own machines, this face, this material. The worked example is only an example: replace every figure with yours.

Circuit: _____ **Date:** _____ **Shift (day or night):** _____

Step	Calculation	Your figure
1 · Shift volume	From the weekly plan, adjusted for today's machines: bcm or t	
2 · Loads	Volume ÷ payload per load = loads	
3 · Digging hours	Shift hours − pre-start − travel − crib − fuel − shift change = digging hours	
4 · Loads an hour	Loads ÷ digging hours = loads an hour; ÷ trucks = per truck per hour	
5 · Buckets a load	Buckets that fill a truck on this face, with this machine, in this material	
6 · Buckets a minute, seconds a bucket	Loads an hour × buckets a load ÷ 60 = buckets a minute; 60 ÷ buckets a minute = seconds a bucket	

Worked example: 1 · Shift volume: 12,000 bcm · 2 · Loads: 12,000 ÷ 60 = 200 loads · 3 · Digging hours: 12 − 3 (pre-start and travel, crib, fuel, shift change) = 9 h · 4 · Loads an hour: 200 ÷ 9 = about 22 an hour; ÷ 4 trucks = about 5.5 per truck per hour · 5 · Buckets a load: 5 · 6 · Buckets a minute, seconds a bucket: 22.2 × 5 ÷ 60 = about 1.85 a minute; 60 ÷ 1.85 = about 32 s

Say it in one sentence: "We need ☐ loads this shift: that is ☐ an hour, ☐ per truck per hour, and a bucket every ☐ seconds."

First bucket (time): _____ **Last bucket (time):** _____

Loads target, block by block: Block 1 ☐ loads · Block 2 ☐ loads · Block 3 ☐ loads · Block 4 ☐ loads · Shift ☐ loads

The number you commit is in loads. You report it in loads and buckets per hour. Buckets a minute and seconds a bucket are only for checking at the face.

Tool 2. The operator's count card

Every block

One card per operator, per shift. The operator writes their own numbers, by hand, at the end of each block. You check them with the operator, in person.

A digger operator counts loads and buckets. A truck operator counts loads and writes the payload, if the truck shows it.

Operator: _____ **Machine:** _____ **Area:** _____

Date: _____ **Shift (day or night):** _____

My target: ☐ **loads this shift ·** ☐ **loads an hour**

Block	Loads (loads)	Buckets (digger only, buckets)	Payload (truck only, bcm or t)	Delays: cause and minutes	Checked (initials)
1					☐
2					☐
3					☐
4					☐
Shift					☐

First bucket and last bucket (times, digger): _____

What got in my way this shift: _____

Tool 3. The operator scorecard

Each roster, on the wall

Keep it up to date from the count cards, on the wall in the pre-start room, where everyone can see it. It shows each operator's trend over the roster, not a comparison with the last shift. At least once a roster, name and congratulate the top performers in front of the crew. Never call out poor performers in front of the crew.

Crew: _____ **Roster:** _____ **Target:** ☐ loads an hour (digger) · ☐ loads per truck per hour

Operator	Machine	Shifts this roster (shifts)	Loads an hour, roster trend (loads an hour)	Against target (%)	First bucket on time (shifts)	Top performer
						☐
						☐
						☐
						☐
						☐
						☐

Named and congratulated at the pre-start on (date): _____

Tool 4. The handover sheet and cards

Both handovers: 05:00 and 17:00

Off-going supervisor: _____ Oncoming supervisor: _____ Date: _____ Shift (day or night): _____

Last shift, in PAVA (from the report)

Measure	Plan	Actual	Variance and impact	Action taken
Loads (loads)				
Loads an hour (loads an hour)				
Buckets an hour (buckets an hour)				
First and last bucket (times)				
Top delay (minutes)				

Set up, and to set up

Item	What, and where	Name
What the last shift left set up for me (or did not)		
What I must leave set up for the next shift		
The focus or priority areas, and the contingencies in force		
Things escalated, still to be resolved		
Key actions for the crew, written on the plan for the pre-start		
Pit plan checked against the mud map; hazards, dig position, floor and roads		

Coaching card: the handover

#	What good looks like	Yes / No
1	Started and finished on time, with all supervisors and Superintendents there; meeting behaviours kept.	☐ Yes ☐ No
2	The plan for this shift was ready before 05:00 and used in the handover.	☐ Yes ☐ No
3	Last shift taken over in PAVA from the report: dig position, rates, hazards, what was escalated, work not finished.	☐ Yes ☐ No
4	Pit plan and layout checked in the office against the mud map or pit map, area by area; understanding checked.	☐ Yes ☐ No
5	Targets set for the staff and gear really there this shift, not the budget or an average.	☐ Yes ☐ No
6	The focus or priority areas for the shift agreed.	☐ Yes ☐ No
7	The contingencies in force reviewed, with the TARPs and golden rules; each contingency has a name.	☐ Yes ☐ No
8	The next 24 hours checked across drill, blast, load and haul, ROM and maintenance: services, supplies, gear and order of work.	☐ Yes ☐ No
9	Each risk to the plan either planned around or escalated, with a name on it.	☐ Yes ☐ No
10	Told what this shift must set up for the next. Checked what the last shift left set up. Each item has a name.	☐ Yes ☐ No
11	What did not fit was challenged; changes agreed with the Superintendent before pre-start.	☐ Yes ☐ No
12	Key actions written on the plan as tasks for the crew, ready for pre-start.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Actions required from any No: _____ **Owner:** _____ **By:** _____

Coaching card: the pre-start

#	What good looks like	Yes / No
13	Started and finished on time, with the whole crew there; meeting behaviours kept.	☐ Yes ☐ No
14	Run from the plan made at the handover, checked by the supervisor before the meeting.	☐ Yes ☐ No
15	Safety first: today's hazards, areas and safety focus, in more detail when something has changed or the crew is new.	☐ Yes ☐ No
16	Last shift's results given briefly, area by area.	☐ Yes ☐ No
17	Each area and its key work described; every operator given a machine, an area and a task.	☐ Yes ☐ No
18	The shift target said to the crew in loads and buckets, with first and last bucket times.	☐ Yes ☐ No
19	Operator scorecard up to date and on the wall where all can see it; at least once a roster, the top performers named and congratulated, on the roster's trend.	☐ Yes ☐ No
20	Spoken slowly and clearly, with authority and energy.	☐ Yes ☐ No
21	Understanding checked by name: operators can say first and last bucket, the targets, the contingencies, the priority areas and the risks to the numbers; a safe and productive shift expected.	☐ Yes ☐ No
22	Crew out of the room within 30 seconds of the end, straight to their gear.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Actions required from any No: _____ **Owner:** _____ **By:** _____

Tool 5. The four block cards

Every shift

Each block card is a stand-alone tool: take it into the field. Mark each line Yes or No at the end of the block. Your Superintendent coaches you on it.

Coaching card: Block 1 · Start up

06:15 to 09:00

#	What good looks like	Yes / No
23	First bucket on time; all operators to their gear on time.	☐ Yes ☐ No
24	Go-lines operating efficiently; fuel plan in place and kept to; all work areas inspected and set up for safety.	☐ Yes ☐ No
25	The handover you got verified as correct.	☐ Yes ☐ No
26	The shift target confirmed as achievable with the crew actually on shift, the gear available and the pit conditions as expected.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Supervisor: _____ **Date:** _____ **Shift (day or night):** _____

First bucket (time): _____

Loads: target ☐ **loads / my figure** ☐ **loads / the system** ☐ **loads**

The number I commit at the 09:00 Drive-around (night shift: at the top of the Shift Report): ☐ **loads**

Actions required from any No: _____ **Owner:** _____ **By:** _____

Coaching card: Block 2 · Execution

09:30 to 11:30

#	What good looks like	Yes / No
27	Loads, buckets, payloads from the operators' counts, beside the system's; differences settled.	☐ Yes ☐ No
28	Top delay acted on; one safety interaction at the work.	☐ Yes ☐ No
29	Touch-base 11:30, in numbers: continue, correct, prepare or escalate.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Supervisor: _____ **Date:** _____ **Shift (day or night):** _____

Loads: target ☐ **loads / my figure** ☐ **loads / the system** ☐ **loads**

Buckets: target ☐ **buckets an hour / my figure** ☐ **buckets an hour**

Top delay: the cause _____ **, minutes lost** ☐ **minutes**

Decision at the 11:30 touch-base, for each area (continue, correct, prepare or escalate): _____

Actions required from any No: _____ **Owner:** _____ **By:** _____

Coaching card: Block 3 · Optimise

11:45 to 14:30

#	What good looks like	Yes / No
30	Stopwatch on hang, queue, load and cycle time; what was needed changed and measured.	☐ Yes ☐ No
31	One operator coached at the work.	☐ Yes ☐ No
32	Check-in 14:30: the gap, and the shift left.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Supervisor: _____ Date: _____ Shift (day or night): _____

Three trucks	Truck 1	Truck 2	Truck 3	Target
Hang time (seconds)				
Queue time (seconds)				
Load time (seconds)				
Cycle time (minutes)				

What I changed, and what I expected: _____

The gap at the 14:30 check-in: ☐ loads; shift left: ☐ hours

Actions required from any No: _____ Owner: _____ By: _____

Coaching card: Block 4 · Set up and hand over

14:45 to the end of the shift

#	What good looks like	Yes / No
33	Last bucket protected.	☐ Yes ☐ No
34	Next shift's areas inspected and set up.	☐ Yes ☐ No
35	What we got away with, what went wrong and the decisions to make before the work, written down.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Supervisor: _____ **Date:** _____ **Shift (day or night):** _____

Last bucket (time): _____

Loads, shift total: target ☐ **loads / my figure** ☐ **loads / the system** ☐ **loads**

What we got away with: _____

Decisions to make before the work: _____

Actions required from any No: _____ **Owner:** _____ **By:** _____

Tool 6. The Shift Report

Written through the blocks; done by about 16:00

Write it through the blocks; finish it in Block 4. Every number is in loads and buckets per hour, in PAVA.

Supervisor: _____ **Crew:** _____ **Superintendent:** _____

Date: _____ **Shift (day or night):** _____

The shift target: [ ] **loads · The number I committed at the Drive-around (written before Block 2):** [ ] **loads**

Production by digger

Digger and area	Loads, plan / actual (loads)	Loads an hour, plan / actual	Buckets an hour, plan / actual	Payload, plan / actual (bcm or t)	Digging hours, plan / actual (hours)	Variance and impact	Action taken

Block	Loads plan (loads)	Loads actual, my figure (loads)	The system (loads)	Variance and impact	Action taken
1 · Start up					
2 · Execution					
3 · Optimise					
4 · Set up and hand over					
Shift					

Measure	Plan	Actual	Variance and impact	Action taken
First bucket (time)				
Last bucket (time)				
Cycle time (minutes)				
Trucks on the circuit (trucks)				

Delay	Unit	Minutes	Cause	Action taken

Work areas inspected (areas): _____ · Safety interactions held (interactions): _____

Hazards found, and their controls: _____

What we got away with: _____

The gap, in one line (the number and the cause): _____

Each variance: what could have foreseen or avoided it, and does it reach the next shift: _____

Jobs not finished, and risks to the next shift: _____

Escalated, and still to be resolved: _____

Key actions I need from the next shift: _____

Coaching card: the Shift Report

#	What good looks like	Yes / No
36	Written on the Shift Report sheet through the blocks, from the operators' counts, not from memory.	☐ Yes ☐ No
37	Plan against actual, in PAVA, in loads and buckets per hour, block by block and at the end, with any statutory reports.	☐ Yes ☐ No
38	Jobs not finished this shift are listed.	☐ Yes ☐ No
39	It shows what was done, and when, each time the shift went off plan.	☐ Yes ☐ No
40	The gap in one line; each variance explained (its impact, the action taken and whether it worked) or accepted in writing.	☐ Yes ☐ No
41	What could have foreseen or avoided each variance, and whether it reaches the next shift, written down.	☐ Yes ☐ No
42	Risks to the next shift written down, with the jobs it must plan in.	☐ Yes ☐ No
43	Done by about 16:00, before the push to the last bucket.	☐ Yes ☐ No
44	Explained at the final drive-around and used at the end handover with the oncoming supervisor.	☐ Yes ☐ No
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	☐ Yes ☐ No

Actions required from any No: _____ **Owner:** _____ **By:** _____

The actions page (final drive-around, 16:30 to 16:50)

Variance	Closed, carried or escalated	Owner	By (date)

The coaching point (one): _____ **Checked at the next final drive-around:** ☐ Yes ☐ No

The decision made live, put forward as a rule: _____

The handover to the next crew (Step 5): mark it on the handover card in Tool 4 (lines 1 to 12).

Shift Report signed (supervisor): _____

Received (Superintendent): _____

Tool 7. The assumptions I inherited

Redo each quarter

You did this in Prime. Redo it each quarter, and whenever the ground changes. Ask your Superintendent for the planned, budgeted or priced figures. If nobody can produce them, write that down: it is the first finding. These assumptions become your PADAL numbers (Primary productivity, Annualised work hours, Delay SMU percentage, Ancillary factors, Labour factor): the five numbers your shift is measured on.

Assumption	What was planned (with unit)	What I see over two weeks (same unit)	Matches?	So what for my number?
Digging rate, by digger (loads an hour)			☐ Yes ☐ No	
Trucks on each digger (trucks)			☐ Yes ☐ No	
Truck cycle time (minutes)			☐ Yes ☐ No	
Delay (SMU) allowance (% of engine hours)			☐ Yes ☐ No	
Payload, by truck class (bcm or t)			☐ Yes ☐ No	
Ancillary factors: support gear hours per 1,000 bcm or t (hours)			☐ Yes ☐ No	
Labour factor: people hours per machine engine hour (hours)			☐ Yes ☐ No	
Availability, by fleet (%)			☐ Yes ☐ No	
Haul distance (km)			☐ Yes ☐ No	

Said at the Drive-around on (date): _____ Superintendent's answer: _____

Tool 8. The six behaviours self-evaluation

Each roster

Rate yourself Yes or No for this roster, two ways: what you do on shift (displayed), and what you instil in your crew so it happens when you are not there (instilled). Give one real example behind each Yes. Take the card to your Superintendent; agree one behaviour to work on, and what they will look for in the field. Both sign.

#	Behaviour	Do I...?	Displayed	Instilled	Example behind each Yes
1	Plan	Do I know the target, the risks and the controls before the shift starts?	☐ Yes ☐ No	☐ Yes ☐ No	
2	Assign	Does every person leave pre-start knowing their task, machine, area and target?	☐ Yes ☐ No	☐ Yes ☐ No	
3	Watch	Do I see the work with my own eyes, in the field, every block?	☐ Yes ☐ No	☐ Yes ☐ No	
4	Respond	Do I tell people straight away what was good and what must change, and help them do it better?	☐ Yes ☐ No	☐ Yes ☐ No	
5	Fix	Do I find and remove the cause of a problem, not just the symptom?	☐ Yes ☐ No	☐ Yes ☐ No	
6	Report	Do I report my shift in PAVA (Plan, Actual, Variance/Impact, Actions taken) and set up the next shift for success?	☐ Yes ☐ No	☐ Yes ☐ No	

Name: _____ **Roster:** _____

The one behaviour I will work on this roster: _____

What my Superintendent will look for in the field: _____

Supervisor: **Name** _____ **Signature** _____ **Date** _____

Superintendent: **Name** _____ **Signature** _____ **Date** _____

Tool 9. The fault-tree tool

When a loss keeps coming back

Take the single biggest loss of the period. Walk the tree out loud with your crew or your Superintendent, naming the box at each level. Do not stop at the branch: the branch is the symptom. Keep going until you reach a box you can change, then check the foundation bar.

Fault tree 1 (loading productivity) sits under the rate: tonnes a pass, loading cycle time, truck presentation. Fault tree 2 (digger hours) sits under operating time: secondary work time, utilisation, mechanical availability.

The loss: _____ **Size of the loss:** ☐ **loads or** ☐ **hours**

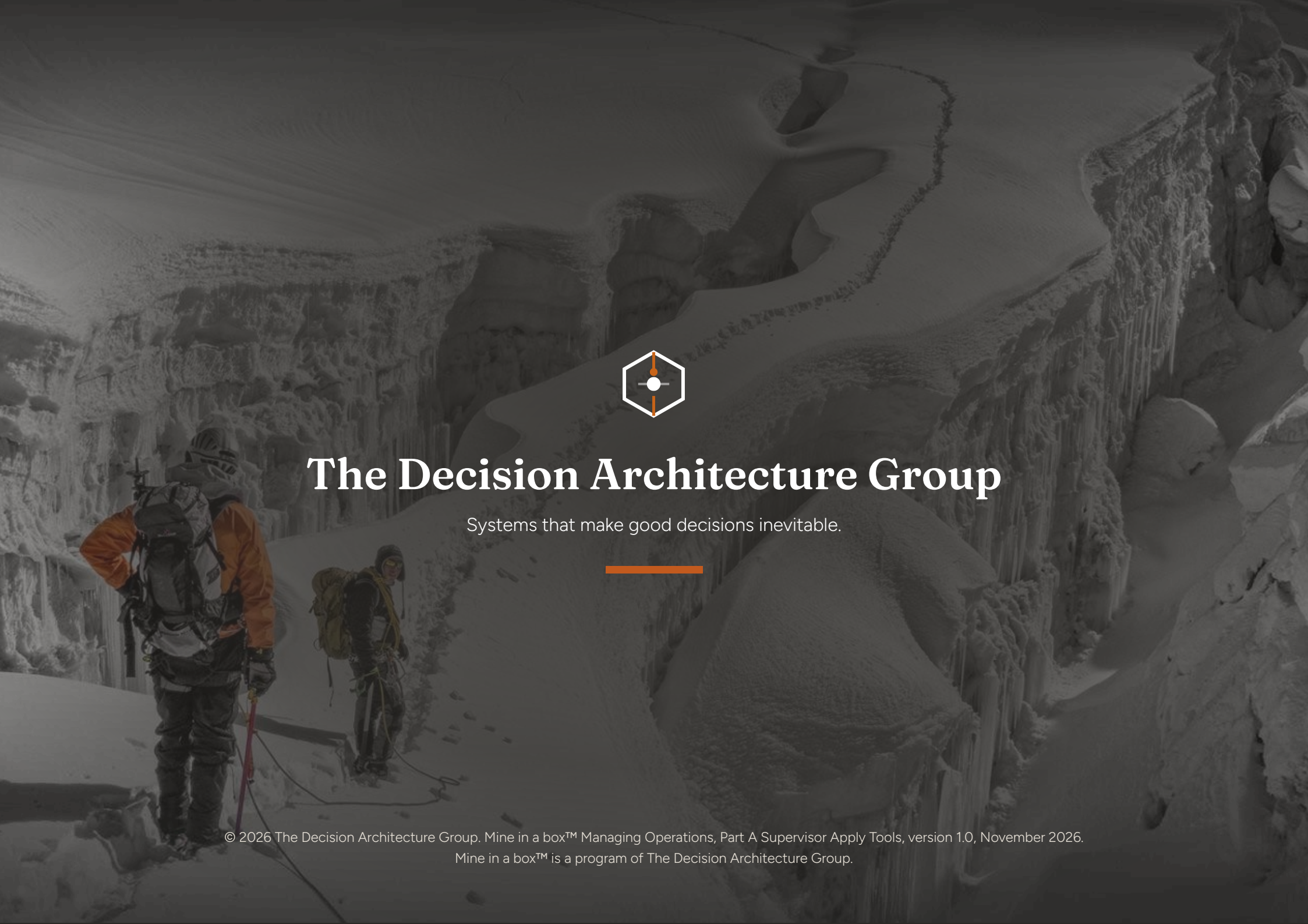
Level	Box I named	Evidence
The lever (operating time or rate)		
The branch		
Second level		
Third level		
Foundation bar (management, standards and procedures, roles and decision rights, competence and training, supervision in the field, culture)		

Where two people named different boxes (a data problem, not a disagreement): _____

The cause (the lowest box reached): _____

The fix, owner and date: _____ **Owner:** _____ **By:** _____

Taken to my Superintendent on (date): _____



The Decision Architecture Group

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